

6G5Z0048_2526_1 Exam 2025/26 Solutions

SECTION A

Answer **ALL** questions in this section.

1. (a) State precisely the definition of the divisibility relation $a|b$ on the integers and prove that the greatest common divisor, d , of any pair of integers, a, b , can be expressed as a linear combination of a and b , i.e.

$$\exists m, n \in \mathbb{Z} \quad d = ma + nb.$$

Hint: Consider the set, S , of all positive linear combinations of a and b , i.e.

$$S = \{xa + yb \mid x, y \in \mathbb{Z} \text{ \& } xa + yb > 0\}.$$

(6 marks)

Solution: We say a divides b , and write $a|b$, if and only if there exists an integer c such that $b = ac$. Consider

$$S = \{xa + yb \mid x, y \in \mathbb{Z} \text{ \& } xa + yb > 0\}.$$

By the well-ordered principle S contains a least element. Call this e , and we have $e = ua + vb$, for some $u, v \in \mathbb{Z}$. We can show that e is a common divisor of a and b . Suppose e does not divide a , then we can divide a as

$$a = qe + r,$$

where the remainder r satisfies $0 < r < e$. Then we can say

$$r = a - qe = a - q(ua + vb) = (1 - qu)a + vb.$$

This shows that $r \in S$, since it is a positive linear combination of a and b . But $r < e$. This contradicts the definition of e as the least element of S . Therefore e divides a . Using a similar argument we can show that e divides b .

So e is a common divisor of a and b , and since $e = ua + vb$, every common divisor of a and b divides e . Therefore e is the greatest common divisor of a and b .

- (b) Use the principle of mathematical induction to prove that

$$\forall n \geq 1 \quad 13 \mid (5^{4n} - 1).$$

You should point out in your argument where you make use of the linear combinations result from part (a) above. (5 marks)

Solution: When $n = 1$, $5^4 - 1 = 624 = 48 \cdot 13$, and clearly $13|624$. So the base case is true. Let us assume that $13|5^{4k} - 1$, for some $k \geq 1$, and then note that

$$\begin{aligned} 5^{4(k+1)} - 1 &= 625 \cdot 5^{4k} - 1 \\ &= 625 \cdot (5^{4k} - 1) + 624. \end{aligned}$$

Note that this last expression is a linear combination of integers divisible by 13, so we have $13|(5^{4(k+1)} - 1)$. Hence, by induction, $13|(5^{4n} - 1)$ for all $n \geq 1$.

(c) Prove that there are infinitely many prime numbers. State clearly any results about divisibility that you rely on. (10 marks)

Solution: We present Euclid's proof by contradiction. Assume, on the contrary, that there are only a finite number of primes, and that they can be listed as $p_1, p_2, \dots, p_n$.

Then consider the integer N defined by

$$N = \left(\prod_{i=1}^n p_i \right) + 1.$$

By the Fundamental theorem of arithmetic N factorizes uniquely into primes, so we can say that there is some j , with $1 \leq j \leq n$ and $p_j|N$.

Rewriting the above equation as

$$1 = N - \left(\prod_{i=1}^n p_i \right),$$

we see that $p_j|1$ as 1 is a linear combination of integers, both divisible by p_j .

But $p_j|1$ contradicts the fact that $p_j > 1$ as p_j is prime.

So we conclude that there are infinitely many primes.

(d) Prove that if an integer of the form $2^n - 1$ is prime then it must be the case that the exponent n is also prime. (4 marks)

Solution: We prove the contrapositive.

Assume that n is not a prime, i.e. that $n = rs$ for integers r and s , both strictly greater than 1.

Then we can factorize $2^n - 1$ as

$$\begin{aligned} 2^n - 1 &= 2^{rs} - 1 \\ &= (2^r)^s - (1)^s \\ &= (2^r - 1) \sum_{j=0}^{s-1} (2^{rj}) \end{aligned}$$

The conditions on r, s imply that this is a genuine factorization, i.e. $1 < 2^r - 1 < 2^n + 1$, and hence $2^n - 1$ is not prime.

SECTION A

2. (a) Let G be a non-empty set and $*$ a binary operation on G , i.e.

$$\forall g_1, g_2 \in G \quad g_1 * g_2 \in G.$$

State the three extra conditions that the pair $(G, *)$ needs to satisfy in order to be called a group and explain their meaning. Illustrate each condition with an example drawn from the group $(\text{GL}(2, \mathbb{R}), \times)$. (8 marks)

Solution: The operation is associative on G , i.e. for all $g, h, k \in G$,

$$g * (h * k) = (g * h) * k.$$

This means that no brackets are needed in products of group elements. Multiplication of matrices, the operation in $\text{GL}(2, \mathbb{R})$, is known to be associative, e.g. for all such matrices A, B, C we have $A(BC) = (AB)C$.

There exists an identity element $e \in G$ satisfying for all $g \in G$, $e * g = g * e = g$. When multiplied by any element the identity leaves the other element unchanged. In $\text{GL}(n, \mathbb{R})$ the identity is $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ as for all 2×2 matrices A , we have $IA = AI = A$.

For all elements $g \in G$ there exists an inverse element $g^{-1} \in G$ satisfying $g * g^{-1} = g^{-1} * g = e$, the identity element of G . In $\text{GL}(n, \mathbb{R})$, the inverse elements are the *inverse* matrices in the linear algebra sense. The inverse of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the matrix $\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

- (b) The external direct product, $\mathbb{Z}_2 \times \mathbb{Z}_2$, is the group consisting of the four ordered pairs, $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$, of elements of $\mathbb{Z}_2$, under the usual operation in an external direct product.

- (i) Clearly define the operation in an external direct product and write down the Cayley table for the group $\mathbb{Z}_2 \times \mathbb{Z}_2$. Also write down the Cayley table for the group $\mathbb{Z}_4$, the integers under addition modulo 4. (3 marks)
- (ii) From the two Cayley tables point out one feature that shows these two groups have a different structure. (2 marks)

Solution:

- (i) Given two groups $(G, *)$ and $(H, \circ)$, their external direct product, $G \times H$, is the group formed by the ordered pairs

$$G \times H = \{(g, h) : g \in G, h \in H\},$$

under the operation defined by

$$(g_1, h_1)(g_2, h_2) = (g_1 * g_2, h_1 \circ h_2).$$

The Cayley tables are

$\mathbb{Z}_2 \times \mathbb{Z}_2$	(0, 0)	(0, 1)	(1, 0)	(1, 1)
(0, 0)	(0, 0)	(0, 1)	(1, 0)	(1, 1)
(0, 1)	(0, 1)	(0, 0)	(1, 1)	(1, 0)
(1, 0)	(1, 0)	(1, 1)	(0, 0)	(0, 1)
(1, 1)	(1, 1)	(1, 0)	(0, 1)	(0, 0)

and

$\mathbb{Z}_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

- (ii) In $\mathbb{Z}_2 \times \mathbb{Z}_2$ every element x satisfies $x + x = e$, i.e. has order 1 or 2. In $\mathbb{Z}_4$ this is not so.

- (c) Which matrices are elements of the group $GL(n, \mathbb{Q})$? Prove that this is a group under the operation of matrix multiplication. You can make use of any facts about matrices from linear algebra, provided you clearly state these in your answer. *Note: the field of coefficients is $\mathbb{Q}$, not $\mathbb{R}$.* (7 marks)

Solution: $GL(n, \mathbb{Q})$ is the group of $n \times n$ matrices with rational coefficients and non-zero determinant under the operation of multiplication. The determinant satisfies

$$\det(AB) = \det(A) \det(B).$$

So if $\det(A)$ and $\det(B)$ are non-zero then so is $\det(AB)$.

Secondly, since matrix multiplication just consists of repeated multiplications and additions involving the entries of the two matrices, the entries of the product matrix will again all be rational numbers.

Therefore $\text{GL}(n, \mathbb{Q})$ is closed under matrix multiplication. Matrix multiplication is known to be associative, i.e. for all $n \times n$ matrices A, B, C ,

$$A(BC) = (AB)C.$$

The identity element is the similarly named $n \times n$ *identity* matrix

$$I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix},$$

which satisfies for all $n \times n$ matrices A , $AI = IA = A$. The entries of I are clearly rational, and it has determinant 1. Hence $I \in \text{GL}(n, \mathbb{Q})$.

Lastly, every $n \times n$ matrix A with non-zero determinant has an inverse (w.r.t. mat. mult.) which has determinant $1/\det(A)$, which is also non-zero. The co-factor formula that gives the inverse again just involves many sums and products of the entries, and so also preserves the rational nature of the entries. Hence A^{-1} is an element of $\text{GL}(n, \mathbb{Q})$.

(d) Consider the set of 2×2 matrices $H \subset \text{GL}(2, \mathbb{Q})$ given by

$$H = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{Q} \text{ \& } ac \neq 0 \right\}.$$

Prove that H forms a subgroup of $\text{GL}(2, \mathbb{Q})$.

(5 marks)

Solution: We will show that the conditions of the subgroup theorem are satisfied by H .

Note that the matrix in H shown in the definition has determinant equal to $ac \neq 0$ so H is indeed a subset of $\text{GL}(2, \mathbb{Q})$. The identity matrix is clearly an element of H (use $a = c = 1, b = 0$). So it just remains to prove that H is closed under matrix multiplication and contains the necessary inverse matrices. Checking closure we find

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & ae + bf \\ 0 & cf \end{pmatrix} \in H.$$

Checking for inverses we find

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \in H.$$

END OF SECTION A

SECTION B

Answer **TWO** questions in this section.

3. (a) State the definition of the congruence relation $a \equiv b \pmod{n}$. How does it relate to the smallest positive remainders left by a and b upon division by n ? (3 marks)

Solution: The definition of $a \equiv b \pmod{n}$ is that $n|(a - b)$.

And $a \equiv b \pmod{n}$ is equivalent to a and b leaving the same small positive remainder, r with $0 \leq r < n$, upon division by n .

- (b) Let $a, n \in \mathbb{Z}$ with $n > 0$. Prove that a multiplicative inverse, a^{-1} , for a modulo n exists if and only if $\gcd(a, n) = 1$. Find the inverse, y , for 13 modulo 100, that lies in the range $1 \leq y \leq 99$. (10 marks)

Solution: We establish a chain of equivalences linking the two conditions.

$$\begin{aligned}\gcd(a, n) = 1 &\Leftrightarrow \exists r, s \in \mathbb{Z} \quad ar + ns = 1, \text{ by the Euclidean algorithm and Bezout's identity} \\ &\Leftrightarrow \exists r, s \in \mathbb{Z} \quad ar - 1 = n(-s) \\ &\Leftrightarrow \exists r \in \mathbb{Z} \quad n|ar - 1, \text{ by the defn. of divisibility} \\ &\Leftrightarrow \exists r \in \mathbb{Z} \quad ar \equiv 1 \pmod{n}, \text{ by the defn. of congruence} \\ &\Leftrightarrow a \text{ is invertible modulo } n\end{aligned}$$

To find 13^{-1} modulo 100 we run the extended Euclidean algorithm on the pair 13, 100 to find:

$$\begin{aligned}100 &= 7 \cdot 13 + 9 \\ 13 &= 1 \cdot 9 + 4 \\ 9 &= 2 \cdot 4 + 1\end{aligned}$$

then back substitution provides:

$$\begin{aligned}1 &= 9 - 2 \cdot 4 \\ &= 9 - 2 \cdot (13 - 9) \\ &= -2 \cdot 13 + 3 \cdot 9 \\ &= -2 \cdot 13 + 3 \cdot (100 - 7 \cdot 13) \\ &= 3 \cdot 100 - 23 \cdot 13.\end{aligned}$$

This shows that -23 is an inverse for 13 modulo 100. The inverse in the required range is thus $y = 77$, as $77 \equiv -23 \pmod{100}$.

- (c) State the definition of the Euler totient function ϕ and prove that for any prime p and positive integer n , that ϕ satisfies

$$\phi(p^n) = p^{n-1}(p - 1).$$

(7 marks)

Solution: The definition of $\phi(m)$ is the number of integers j such that $1 \leq j \leq m$ and $\gcd(j, m) = 1$.

Of the integers j in the range $1 \leq j \leq p^n$, the ones that are not coprime to p^n are those integers j such that $p|j$. This is because p is a prime so the only divisors of p^n are other powers of p .

Within the sequence $1, 2, 3, \dots, p^n$ the multiples of p are

$$p, 2p, 3p, \dots, p^{n-1}p = p^n,$$

so there are p^{n-1} of these, and so the number of j that **are** coprime to p^n is

$$\begin{aligned}\phi(p^n) &= p^n - p^{n-1} \\ &= p^{n-1}(p - 1),\end{aligned}$$

as required.

- (d) Write down the multiplicative formula for $\phi(n)$ that gives its value based on the prime factorization of n , where $n \in \mathbb{Z}^+$. Use the formula to calculate $\phi(100)$. (5 marks)

Solution: The full formula for ϕ is

$$\phi(n) = \prod_{i=1}^r (p_i^{a_i-1}(p_i - 1)),$$

where n has the prime factorization $n = \prod_{i=1}^r p_i^{a_i}$, for distinct primes p_i and non-zero exponents a_i .

We calculate $\phi(100)$ using this and the prime factorization of 100.

$$\begin{aligned}\phi(100) &= \phi(2^2 \cdot 5^2) \\ &= 2^1 \cdot (2 - 1) \cdot 5^1 \cdot (5 - 1) \\ &= 2 \cdot 5 \cdot 4 \\ &= 40\end{aligned}$$

SECTION B

4. (a) State Lagrange's theorem on the orders of subgroups of a finite group G . (3 marks)

Solution: Anything equivalent to: If G is a finite group and H a subgroup of G then $|H|$ divides $|G|$.

- (b) Let H be a subgroup of a finite group G .

- (i) State the definition of the left and right cosets of H in G . (2 marks)
- (ii) Let $g_1, g_2 \in G$. Prove that the left-cosets g_1H and g_2H are either equal or disjoint, i.e.

$$g_1H = g_2H \quad \text{or} \quad g_1H \cap g_2H = \emptyset.$$

(6 marks)

- (iii) Prove that all cosets of H in G contain the same number of elements. (3 marks)
- (iv) Then show how parts (ii) and (iii) above can be used to prove Lagrange's theorem. (4 marks)

Solution:

- (i) The left-coset of H in G with representative $g \in G$ is the subset gH of G defined by

$$gH = \{gh : h \in H\}.$$

The right-coset of H in G with representative $g \in G$ is the subset Hg of G defined by

$$Hg = \{hg : h \in H\}.$$

- (ii) Suppose that $x \in g_1H \cap g_2H$. Therefore there exists $h_1, h_2 \in H$ such that $x = g_1h_1 = g_2h_2$ which implies that $g_1 = g_2h_2h_1^{-1}$ and $g_2 = g_1h_1h_2^{-1}$. Then for all $h \in H$ we have

$$g_1h = g_2h_2h_1^{-1}h \in g_2H,$$

as the product $h_2h_1^{-1}h$ is clearly in H as H is a subgroup. So $g_1H \subseteq g_2H$. Similarly, for all $h \in H$ we have

$$g_2h = g_1h_1h_2^{-1}h \in g_1H.$$

So $g_2H \subseteq g_1H$. Therefore $g_1H = g_2H$. This proves that if the intersection of g_1H and g_2H is non-empty (there is such an x as above) then the two cosets are equal.

So in conclusion any two cosets are either disjoint or equal.

- (iii) We will show that all cosets of H contain $|H|$ elements by exhibiting the bijection $\phi : H \rightarrow gH$ defined by $\phi(h) = gh$.

That this is surjective is clear from the definition of gH .

Suppose that $\phi(h_1) = \phi(h_2)$, i.e. $gh_1 = gh_2$. Applying the group cancellation law (multiplying on left by g^{-1}) shows that $h_1 = h_2$. So ϕ is also injective, and hence a bijection. Therefore all cosets contain the same number, $|H|$, of elements.

- (iv) The collection of cosets of H in G forms a partition of G , meaning every element of G is contained in some coset, for if $g \in G$ then $g \in gH$ as $g = ge \in gH$ as $e \in H$. Also different subsets of the partition are disjoint (from part (ii)).

So the total number of elements of G can be found by summing the number of elements in each distinct coset. Part (iii) shows that every coset has the same number of elements, namely $|H|$.

So we get the result of Lagrange's theorem that

$$|G| = [G : H]|H|,$$

where $[G : H]$, the index of H in G is the number of distinct cosets of H in G .

- (c) The dihedral group D_8 is generated by the pair of elements r, s which are subject to the relations $r^8 = e$, $s^2 = e$ and $sr = r^{-1}s$. Consider the subgroup H of D_8 given by

$$H = \{e, r^2, r^4, r^6\}.$$

- (i) Work out the elements of each left coset of H in D_8 . (4 marks)
- (ii) Give an example of a subgroup K of D_8 and an element $x \in D_8$ for which

$$xK \neq Kx.$$

(3 marks)

Solution:

(i) The distinct cosets are (one on each line)

$$\begin{aligned}H &= r^2H = r^4H = r^6H = \{e, r^2, r^4, r^6\} \\rH &= r^3H = r^5H = r^7H = \{r, r^3, r^5, r^7\} \\sH &= r^2sH = r^4sH = r^6sH = \{s, r^2s, r^4s, r^6s\} \\rsH &= r^3sH = r^5sH = r^7sH = \{rs, r^3s, r^5s, r^7s\}.\end{aligned}$$

(ii) An example is $K = \{e, s\}$ and $x = r$. For then we have

$$rK = \{r, rs\} \text{ and } Kr = \{r, r^7s\},$$

which are different since $rs \neq r^7s$.

SECTION B

5. (a) Consider the congruence

$$21x \equiv 14 \pmod{49}.$$

User relevant result(s) from the theory of congruences to find all the solutions. (9 marks)

Solution: Using the result about linear congruences we see that $\gcd(21, 49) = 7$ and $7 \mid 14$ so there are seven solutions to the congruence given by

$$t + 7i, \quad (i = 0, 1, 2, 3, 4, 5, 6)$$

where t is the unique solution $0 \leq t \leq 6$ to the reduced congruence

$$3x \equiv 2 \pmod{7}.$$

From the Euclidean algorithm (or simple observation) we see that $3^{-1} \equiv 5 \pmod{7}$ so that

$$t \equiv 5 \times 2 \equiv 10 \equiv 3 \pmod{7}.$$

So the seven solutions to the original congruence are given by

$$x \equiv 3, 10, 17, 24, 31, 38, 45 \pmod{49}.$$

(b) Use the Chinese Remainder Theorem to describe the integers x that satisfy all three of the following congruences simultaneously,

$$x \equiv 4 \pmod{7}$$

$$x \equiv 3 \pmod{13}$$

$$x \equiv 9 \pmod{17}.$$

Your final answer should be in the form of a single congruence class for x modulo an appropriate modulus. (10 marks)

Solution: We use the Chinese Remainder theorem from the handout. Firstly the three moduli $m_1 = 7$, $m_2 = 13$, $m_3 = 17$ are distinct primes, hence pairwise coprime so the theorem applies.

Let $M = 7 \times 13 \times 17 = 1547$. Then we have

$$M_1 = 221, M_2 = 119, M_3 = 91,$$

and the multiplicative inverses M'_i of M_i modulo m_i are obtained from the following:

$$\begin{aligned} M'_1 &\equiv (221)^{-1} \pmod{7} \\ &\equiv 4^{-1} \pmod{7} \\ &\equiv 2, \pmod{7} \end{aligned}$$

and then

$$\begin{aligned} M'_2 &\equiv (119)^{-1} \pmod{13} \\ &\equiv 2^{-1} \pmod{13} \\ &\equiv 7, \pmod{13} \end{aligned}$$

and then

$$\begin{aligned} M'_3 &\equiv (91)^{-1} \pmod{17} \\ &\equiv 6^{-1} \pmod{17} \\ &\equiv 3, \pmod{19} \end{aligned}$$

The solutions x to the system of congruences in the question are all of the integers in the congruence class

$$\begin{aligned} x &\equiv 4M_1M'_1 + 3M_2M'_2 + 9M_3M'_3 \pmod{1547} \\ &\equiv 4 \cdot 221 \cdot 2 + 3 \cdot 119 \cdot 7 + 9 \cdot 91 \cdot 3 \pmod{1547} \\ &\equiv 6724 \equiv 536 \pmod{1547}. \end{aligned}$$

- (c) Use the Legendre symbol, the law of quadratic reciprocity and other relevant properties to show that there are integer solutions to the congruence

$$x^2 \equiv 307 \pmod{601}.$$

(You can use the fact that 307 and 601 are both prime.)

(6 marks)

Solution: Solutions x exist if and only if 307 is a quadratic residue modulo 601, i.e. iff the Legendre symbol satisfies $(307|601) = +1$.

We evaluate the Legendre symbol as follows:

$$\begin{aligned} (307|601) &= (601|307), \quad \text{quadratic reciprocity, } 601 \equiv 1 \pmod{4} \\ &= (294|307), \quad \text{since } 601 \equiv 294 \pmod{307} \end{aligned}$$

$$\begin{aligned}
&= (2 \cdot 3 \cdot 7^2 | 307), \quad \text{factorizing 294} \\
&= (2 | 307) \cdot (3 | 307) \cdot (7 | 307)^2 \quad \text{by the mult. prop. of the Legendre symbol} \\
&= (2 | 307) \cdot (3 | 307), \quad \text{since any Legendre symbol is } \pm 1 \\
&= (307 | 3), \quad \text{quad. recip., } 307 \equiv 3 \pmod{4}, (2 | 307) = -1 \text{ since } 307 \equiv 3 \pmod{8} \\
&= (1 | 3), \quad \text{since } 307 \equiv 1 \pmod{3} \\
&= 1, \quad \text{as 1 is always a quad. residue}
\end{aligned}$$

Thus, $(307 | 601) = +1$, and therefore 307 is a quadratic residue modulo 601, and so solutions x exists to the congruence in question.

SECTION B

6. (a) Give the definition of a **normal subgroup**. (3 marks)

Solution: A subgroup H of a group G is normal in G iff for all $g \in G$ we have $gH = Hg$, i.e. left and right cosets of H in G , represented by the same element g , are equal.

- (b) Prove that a subgroup N of a group G is normal in G if and only if

$$\forall g \in G \quad gNg^{-1} \subset N.$$

(5 marks)

Solution: Suppose N is normal in G . Then we have

$$\begin{aligned} & \forall g \in G \quad gN = Ng \\ \Rightarrow & \forall g \in G \forall n \in N \exists m \in N \quad gn = mg \\ \Rightarrow & \forall g \in G \forall n \in N \exists m \in N \quad gng^{-1} = m \\ \Rightarrow & \forall g \in G \forall n \in N \quad gng^{-1} \in N \\ \Rightarrow & \forall g \in G \quad gNg^{-1} \subset N \end{aligned}$$

Conversely, suppose that $\forall g \in G \quad gNg^{-1} \subset N$. This is equivalent to

$$\forall g \in G \forall n \in N \exists m \in N \quad gng^{-1} = m. \quad (*)$$

Consider any $gn \in gN$. From $(*)$ we know $gng^{-1} = m \in N$ for some $m \in N$, hence $gn = mg \in Ng$. Therefore $gN \subset Ng$. Now consider any $ng \in Ng$. From $(*)$ (but applied to g^{-1} rather than g) we know that $g^{-1}ng = m \in N$ for some $m \in N$, hence $ng = gm \in gN$. Therefore $Ng \subset gN$, and so $gN = Ng$, as required.

- (c) If G is a group and $g \in G$ then the centralizer, $C(g)$, of g in G is the subgroup of G defined by

$$C(g) = \{h \in G \mid hg = gh\},$$

i.e. all the elements from G that commute with g .

Prove that if g generates a normal subgroup of G , i.e. $\langle g \rangle$ is normal in G , then the centralizer $C(g)$ is also normal in G . (9 marks)

Solution: Let $g \in G$. Suppose that $\langle g \rangle$ is normal in G , so by part (c) we have $\forall h \in G \ h \langle g \rangle h^{-1} \subset \langle g \rangle$. This means that

$$\forall h \in G \ \forall n \in \mathbb{Z} \ \exists m \in \mathbb{Z} \ h g^n h^{-1} = g^m. \quad (*)$$

To prove that $C(g)$ is normal in G we need to show that for all $x \in G$ we have $x C(g) x^{-1} \subset C(g)$. So let $x \in G$ and let $h \in C(g)$. We now need to show that $x h x^{-1}$ is in $C(g)$, i.e. that it commutes with g . This is proved by the following:

$$\begin{aligned} x h x^{-1} g &= x h g^m x^{-1}, \text{ since } x^{-1} g = g^m x^{-1}, \text{ for some } m \in \mathbb{Z}, \text{ by } (*) \\ &= x g^m h x^{-1}, \text{ since } h \text{ commutes with } g \text{ as } h \in C(g) \\ &= g x h x^{-1}. \text{ as } x^{-1} g = g^m x^{-1} \Leftrightarrow g x = x g^m \end{aligned}$$

This chain of equations shows that $x h x^{-1}$ commutes with g , as required. Therefore $C(g)$ is indeed normal in G .

(d) Let $x \in \mathbb{R}$ with $x \neq 0, 1$ and let X be the matrix

$$X = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix},$$

considered as an element of the group $\text{GL}(2, \mathbb{R})$. Identify exactly which matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belong to the centralizer $C(X)$. Your conclusion should be given in terms of conditions on the entries a, b, c, d .

(8 marks)

Solution: A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is in $C(X)$ iff it commutes with X . Examining this condition we find:

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} ax & b \\ cx & d \end{pmatrix} &= \begin{pmatrix} ax & bx \\ c & d \end{pmatrix} \\ \Leftrightarrow b &= bx \text{ and } c = cx \\ \Leftrightarrow b &= 0 \text{ and } c = 0, \text{ since } x \neq 1 \end{aligned}$$

From this we see that

$$C(X) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbb{R}) : b = c = 0 \right\},$$

i.e. the commutator of X consists of all the diagonal matrices.

END OF SECTION B
END OF QUESTIONS